MENU-DRIVEN C PROGRAM

Date Format Conversion and
Printing All Sub-strings of a String

SUBMITTED BY

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1. Problem Statement

Write a menu-driven C program to perform the following operations using separate functions:

- Change the date format from **dd/mm/yyyy** to **dd-mmm-yy**. Example: Input: 12/10/2026, Output: 12-Oct-26.
- Print all sub-strings of a string. Example: Input: abc, Output: a,ab,abc,b,bc,c.

2. Program Design

The program is divided into small functions so that each function performs exactly one job. The menu is shown inside a do-while loop and a switch statement calls the function for the chosen option. The loop stops only when the user selects Exit.

Function	Purpose
`main()`	Displays the menu repeatedly, reads the user's choice and calls the required function.
`displayMenu()`	Prints the menu options on the screen.
`isLeapYear()`	Returns 1 if the given year is a leap year, otherwise 0.
`isValidDate()`	Checks whether the day, month and year together form a real calendar date.
`convertDate()`	Converts a date from dd/mm/yyyy to dd-mmm-yy. Returns 0 if the date is invalid.
`printSubstrings()`	Prints all sub-strings of the given string, separated by commas.
`clearInputBuffer()`	Removes leftover characters from the input buffer after scanf().

3. Algorithms

3.1 Changing the date format (convertDate)

1. Read the date as a string in *dd/mm/yyyy* format.
2. Check that the string has exactly 10 characters, that the 3rd and 6th characters are '/', and that every other character is a digit.
3. Extract the day, month and year as integers using sscanf().
4. Validate the date using isValidDate(): the month must be 1 to 12 and the day must not exceed the number of days in that month. February has 29 days in a leap year.
5. Pick the three-letter month name (Jan to Dec) from an array of month names.
6. Take the last two digits of the year using year % 100.
7. Build the result with sprintf() using the format "%02d-%s-%02d" and display it.

A year is a leap year if it is divisible by 4 and not by 100, or if it is divisible by 400.

3.2 Printing all sub-strings (printSubstrings)

A sub-string is a continuous part of a string. It is fixed by its starting position i and its ending position j, where $i \leq j$. The function uses three nested loops:

1. Find the length of the string using strlen().
2. The outer loop (i) runs from 0 to length − 1 and selects the starting position.
3. The middle loop (j) runs from i to length − 1 and selects the ending position.
4. The inner loop (k) runs from i to j and prints the characters of the sub-string.
5. A flag variable is used so that a comma is printed before every sub-string except the first one.
6. Print a new line at the end.

Working for the string "abc":

Start (i)	Character	Sub-strings printed
0	a	a, ab, abc
1	b	b, bc
2	c	c

A string of length n has $n(n + 1)/2$ sub-strings, so "abc" has $3 \times 4 / 2 = 6$ sub-strings.

4. Source Code

File name: `menu_program.c`

```
/*
 * Program : Menu-driven program for date format conversion
 *           and printing all sub-strings of a string
 * Name    : Rudra Pratap Singh
 * SAP ID  : 590032811
 * Batch   : 16
 */

#include <stdio.h>
#include <string.h>

#define MAX 100

/* Function prototypes */
void displayMenu(void);
int  isLeapYear(int year);
int  isValidDate(int day, int month, int year);
int  convertDate(char date[], char result[]);
void printSubstrings(char str[]);
void clearInputBuffer(void);

int main(void)
{
    int  choice;
    char date[MAX];
    char newDate[MAX];
    char str[MAX];
```

```
    do
    {
        displayMenu();

        if (scanf("%d", &choice) != 1)
        {
            choice = 0;               /* non-numeric input */
        }
        clearInputBuffer();

        switch (choice)
        {
            case 1:
                printf("Enter date (dd/mm/yyyy): ");
                scanf("%s", date);
                clearInputBuffer();

                if (convertDate(date, newDate))
                {
                    printf("Output: %s\n", newDate);
                }
                else
                {
                    printf("Invalid date! Please use dd/mm/yyyy format.\n");
                }
                break;

            case 2:
                printf("Enter a string: ");
                scanf("%[^\n]", str);
                clearInputBuffer();

                printf("Output: ");
                printSubstrings(str);
                break;

            case 3:
                printf("Exiting the program. Thank you!\n");
                break;

            default:
                printf("Invalid choice! Please enter 1, 2 or 3.\n");
        }

        printf("\n");

    } while (choice != 3);

    return 0;
}

/* Displays the menu */
void displayMenu(void)
{
    printf("================================================\n");
    printf("                  MAIN MENU\n");
    printf("================================================\n");
    printf(" 1. Change date format (dd/mm/yyyy to dd-mmm-yy)\n");
    printf(" 2. Print all sub-strings of a string\n");
    printf(" 3. Exit\n");
    printf("================================================\n");
    printf("Enter your choice (1-3): ");
}

/* Returns 1 if the year is a leap year, otherwise 0 */
```

```
int isLeapYear(int year)
{
    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
    {
        return 1;
    }
    return 0;
}

/* Returns 1 if day, month and year form a valid date, otherwise 0 */
int isValidDate(int day, int month, int year)
{
    int daysInMonth[] = {31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31};

    if (year < 1 || month < 1 || month > 12 || day < 1)
    {
        return 0;
    }

    if (month == 2 && isLeapYear(year))
    {
        daysInMonth[1] = 29;
    }

    if (day > daysInMonth[month - 1])
    {
        return 0;
    }
    return 1;
}

/*
 * Converts a date from dd/mm/yyyy to dd-mmm-yy.
 * Returns 1 on success and 0 if the input date is invalid.
 */
int convertDate(char date[], char result[])
{
    char *months[] = {"Jan", "Feb", "Mar", "Apr", "May", "Jun",
                      "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"};
    int day, month, year, i;

    /* Format check: exactly 10 characters, '/' at positions 2 and 5 */
    if (strlen(date) != 10 || date[2] != '/' || date[5] != '/')
    {
        return 0;
    }

    /* All other characters must be digits */
    for (i = 0; i < 10; i++)
    {
        if (i != 2 && i != 5 && (date[i] < '0' || date[i] > '9'))
        {
            return 0;
        }
    }

    sscanf(date, "%d/%d/%d", &day, &month, &year);

    if (!isValidDate(day, month, year))
    {
        return 0;
    }

    sprintf(result, "%02d-%s-%02d", day, months[month - 1], year % 100);
    return 1;
```

```
}

/* Prints all sub-strings of str separated by commas */
void printSubstrings(char str[])
{
    int len = strlen(str);
    int i, j, k;
    int first = 1;                    /* controls comma printing */

    for (i = 0; i < len; i++)         /* starting position */
    {
        for (j = i; j < len; j++)     /* ending position */
        {
            if (!first)
            {
                printf(",");
            }
            first = 0;

            for (k = i; k <= j; k++)
            {
                printf("%c", str[k]);
            }
        }
    }
    printf("\n");
}

/* Clears leftover characters (including newline) from the input buffer */
void clearInputBuffer(void)
{
    int ch;
    while ((ch = getchar()) != '\n' && ch != EOF)
    {
    }
}
```

5. Test Cases

The program was compiled with gcc using the -Wall -Wextra options (no warnings) and tested with the following inputs.

No.	Test Description	Input	Output	Result
1	Valid date (example from question)	`12/10/2026`	`12-Oct-26`	**Pass**
2	Single-digit day and month, year 2000	`01/01/2000`	`01-Jan-00`	**Pass**
3	Last day of the year	`31/12/1999`	`31-Dec-99`	**Pass**
4	Leap year date	`29/02/2024`	`29-Feb-24`	**Pass**
5	29 Feb in a non-leap year	`29/02/2023`	`Invalid date message`	**Pass**
6	April has only 30 days	`31/04/2025`	`Invalid date message`	**Pass**
7	Wrong separator	`12-10-2026`	`Invalid date message`	**Pass**
8	String from the question	`abc`	`a,ab,abc,b,bc,c`	**Pass**
9	String of length 4	`abcd`	`a,ab,abc,abcd,b,bc,bcd,c,cd,d`	**Pass**
10	Single-character string	`a`	`a`	**Pass**
11	Menu option out of range	`7`	`Invalid choice message`	**Pass**

In every case the output obtained was the same as the expected output. "Invalid date message" means the program printed: Invalid date! Please use dd/mm/yyyy format.

6. Sample Output

The screens below were captured by running the compiled program. Repeated menus are shortened to one line for readability.

Run 1: Changing the date format

```
================================================
                  MAIN MENU
================================================
 1. Change date format (dd/mm/yyyy to dd-mmm-yy)
 2. Print all sub-strings of a string
 3. Exit
================================================
Enter your choice (1-3): 1
Enter date (dd/mm/yyyy): 12/10/2026
Output: 12-Oct-26
[ main menu is displayed again ]
Enter your choice (1-3): 1
Enter date (dd/mm/yyyy): 29/02/2024
Output: 29-Feb-24
[ main menu is displayed again ]
Enter your choice (1-3): 3
Exiting the program. Thank you!
```

Run 2: Printing all sub-strings

```
================================================
                  MAIN MENU
================================================
 1. Change date format (dd/mm/yyyy to dd-mmm-yy)
 2. Print all sub-strings of a string
 3. Exit
================================================
Enter your choice (1-3): 2
Enter a string: abc
Output: a,ab,abc,b,bc,c
[ main menu is displayed again ]
Enter your choice (1-3): 2
Enter a string: abcd
Output: a,ab,abc,abcd,b,bc,bcd,c,cd,d
[ main menu is displayed again ]
Enter your choice (1-3): 3
Exiting the program. Thank you!
```

Run 3: Invalid inputs

```
================================================
                  MAIN MENU
================================================
 1. Change date format (dd/mm/yyyy to dd-mmm-yy)
 2. Print all sub-strings of a string
 3. Exit
================================================
Enter your choice (1-3): 1
Enter date (dd/mm/yyyy): 31/04/2025
Invalid date! Please use dd/mm/yyyy format.
[ main menu is displayed again ]
Enter your choice (1-3): 1
Enter date (dd/mm/yyyy): 12-10-2026
Invalid date! Please use dd/mm/yyyy format.
[ main menu is displayed again ]
Enter your choice (1-3): 7
Invalid choice! Please enter 1, 2 or 3.
[ main menu is displayed again ]
Enter your choice (1-3): 3
Exiting the program. Thank you!
```

7. Conclusion

The menu-driven program was written in C using a separate function for each task. It converts a date from dd/mm/yyyy to dd-mmm-yy after checking that the date is valid (including leap years), and it prints every sub-string of a given string in the required comma-separated order. Invalid dates, wrong formats and wrong menu choices are handled with proper messages, and the menu keeps running until the user chooses to exit.

Through this program the following concepts were practised: functions and function prototypes, strings and string functions (strlen, sscanf, sprintf), arrays, nested loops, the switch statement and the do-while loop.